Rethinking Multilabel Remote Sensing Image Classification Under Dual-Distribution Instance-Dependent Noise

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***Abstract*—Multilabel scene classification (MLC) is a fundamental task in remote sensing (RS). However, its performance is often limited by noisy annotations, where subtractive noise occurs when inconspicuous targets are missed and additive noise arises when visually confusing categories are falsely added. To simulate such annotation errors, the recently proposed instance-dependent multilabel noise (IDMN) method introduces the first instance-dependent noise generation strategy for RS MLC, driven by the intuition that hard label instances are more susceptible to mislabeling. Specifically, it leverages a single probability distribution of present labels to derive thresholds for both subtractive and additive noise generation. This design determines additive noise by requiring absent-label occurrence probabilities to exceed an irrelevant present-label threshold, leaving its quantity difficult to control and overlooking potentially confusing absent labels. To address this limitation, we propose dual-distribution IDMN (DD-IDMN), which explicitly decouples the score distributions of present and absent labels for calibrated noise generation. Empirical analysis under DD-IDMN reveals that additive noise yields more severe performance degradation than subtractive noise. To combat this DD-IDMN challenge, we propose the prompt-guided activation interference rectification network (PAIR-Net). Through cross-modal attention with learnable text prompts, it learns more discriminative RS feature representations. Furthermore, we introduce an activation-aware dynamic filtering (ADF) module to identify unreliable present labels for suppressing the more destructive additive noise under DD-IDMN. Experiments on UCMerced, AID, and MLRSNet demonstrate that PAIR-Net consistently outperforms representative noise-robust multilabel methods under complex instance-dependent noise. Our code is available at https://github.com/BasilLeave/PAIR-Net**

***Index Terms*—Multilabel noise, multilabel scene classification (MLC), noisy labels, remote sensing (RS).**

I. Introduction

REMOTE sensing (RS) multilabel scene classification (MLC) is an important task in RS image analysis, aiming to assign relevant semantic categories to a given RS image [1], [2], [3]. It supports various applications, such as urban planning [4], [5], [6], environmental monitoring [7], [8], [9], and disaster management [10], [11]. Despite its practical importance, constructing accurately annotated multilabel datasets remains expensive since multiple land-cover categories may coexist within a single image [12], [13], [14]. Consequently, real-world annotations inevitably contain noisy labels and training deep neural networks (DNNs) with such corrupted supervision can lead to severe performance degradation due to the memorization effect [15], [16], [17].

In multilabel classification, label noise is commonly divided into additive, subtractive, and mixed noise. Specifically, additive noise refers to absent categories incorrectly annotated as present, whereas subtractive noise denotes present categories omitted from the annotations. In practice, mixed noise is the most complex scenario, where additive and subtractive errors coexist within the corrupted dataset. Since real-world annotation errors are difficult to obtain and control, existing studies typically inject synthetic noise into clean multilabel RS datasets to evaluate noise-robust learning methods [18], [19], [20], [21]. Early approaches mainly follow instance-independent corruption strategies, where label flipping is independent of image content. For example, uniform noise [18] flips labels with equal probability across predefined categories, while distribution-aware noise [19] assigns category-specific corruption probabilities according to label occurrence frequencies. Although these methods are simple, they ignore the visual characteristics of each image, limiting their ability to capture annotation errors associated with image content in RS images.

To overcome this limitation, recent studies have focused on instance-dependent noise [22], [23], [24], [25], where label corruption depends on image content. In RS multilabel classification, the instance-dependent multilabel noise (IDMN) method [21] is a pioneering noise generation strategy, driven by the intuition that hard labels are more likely to be mislabeled. Benefiting from large-scale image-text pretraining,

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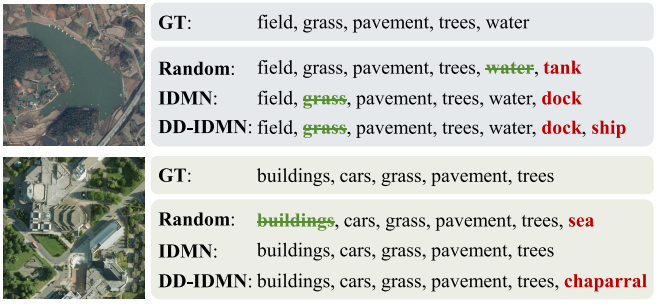


Fig. 1. Comparison of noise injection methods, where green strikethrough and red labels denote subtractive and additive noise, respectively. Random noise flips labels independently of image content. IDMN may miss subtle hard negatives since it selects additive noise only when absent categories show stronger visual evidence than dominant present labels. In contrast, DD-IDMN mines more plausible hard negatives, including white floating objects resembling "ships" and shrub-like regions mistaken for "chaparral."

CLIP [26] has demonstrated superior zero-shot classification capability [27], [28], enabling category relevance estimation from image content for instance-dependent noise modeling. Thus, IDMN employs CLIP to compute a prediction score for each image-category pair, with a higher score indicating a higher estimated occurrence probability. For subtractive noise generation, the key is to identify hard positives and flip their labels from 1 to 0. A hard positive refers to a present category that receives a low prediction score, indicating that it is visually difficult to recognize. In contrast, additive noise generation relies on hard negatives. A hard negative is an absent category with a high prediction score, indicating that it is visually confusing and more likely to be falsely annotated as present. IDMN flips these hard negatives from 0 to 1 when their scores exceed a threshold derived from confident present labels.

However, the additive noise generation mechanism in IDMN exhibits an inherent bias. Intuitively, additive errors mainly arise from visual confusion caused by deceptive localized features, rather than global category saliency. As illustrated in Fig. 1, small white objects floating on water are easily mistakenly annotated as "ship" due to local visual ambiguity, even though they are less conspicuous than the dominant true target "water." Since such localized patterns lack the complete semantic structures of prominent targets (e.g., vast "water" or dense "buildings"), they naturally yield lower CLIP prediction scores than dominant targets. By forcing absent categories to surpass an irrelevant present-label threshold, IDMN may overlook these visually confusing absent labels. Crucially, given the discrepancy between the CLIP prediction distributions of present and absent labels, this cross-distribution thresholding bias makes the generated noise quantity difficult to control, potentially resulting in an unbalanced evaluation benchmark.

To address this limitation, we introduce dual-distribution IDMN (DD-IDMN), which decouples the prediction score distributions of present and absent labels for noise injection. Empirical evaluations reveal that additive noise yields more severe performance degradation than subtractive noise. To combat the DD-IDMN challenge, we propose the prompt-guided activation interference rectification network (PAIR-Net). It introduces cross-modal semantic interaction with learnable text prompts to extract robust feature representations. To suppress the more destructive additive noise observed under DD-IDMN, we further develop an activation-aware dynamic filtering (ADF) module. Since additive noisy labels lack visual instances, their activations tend to be diffuse and are more likely to be contained within activation regions of present categories. Motivated by this observation, we design a spatial containment score (**SCS**) to quantify cross-class semantic interference. It measures the spatial coverage of the target activation region by the union of activation regions from other present labels. A high **SCS** indicates that the label lacks independent visual evidence and is likely to be additive noise. To avoid mistakenly identifying RS categories with natural spatial inclusion relationships as noise, we construct a semantic protection set with an LLM (i.e., ChatGPT [29]). Although the commonly used label co-occurrence matrix [30], [31] can capture statistical label relationships, its estimates are susceptible to noisy training annotations. In contrast, the LLM can effectively leverage superior reasoning ability [32], [33] to identify more stable physical inclusion relationships. During training, PAIR-Net dynamically corrects unreliable positive labels with high **SCS** to mitigate the impact of additive noise. The main contributions are summarized as follows.

1) We identify the cross-distribution thresholding bias in IDMN and propose DD-IDMN. By decoupling the score distributions of present and absent labels, DD-IDMN enables controllable instance-dependent noise modeling and provides a challenging benchmark for evaluating robust RS multilabel classification methods.
2) Empirical analysis under DD-IDMN reveals that additive noise causes more severe degradation than subtractive noise. This finding highlights false positive (FP) suppression as a key robustness challenge beyond instance-independent noisy supervision in RS multilabel classification.
3) We propose PAIR-Net, a label rectification framework for the DD-IDMN challenge. Unlike generic robust learning methods that mainly rely on loss regularization [34] or collaborative training [20], PAIR-Net exploits category-wise visual semantic evidence from activation maps to improve robustness under noisy multilabel supervision.

The remainder of this article is organized as follows. In Section II, we review studies related to our work. In Section III, we present our DD-IDMN and PAIR-Net in detail. Section IV provides comprehensive experimental results and in-depth analyses of PAIR-Net. Section V discusses the limitations. Finally, the conclusion of this work is drawn in Section VI.

II. Related Work

A. RS Scene Classification

Remote sensing (RS) scene classification has received sustained attention, with recent efforts addressing complex RS scene structures [35], [36], [37], scale variation [38], [39], [40], and multimodal observations [41], [42], [43]. Notably,

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RASNet [44] represents each RS image as a continuous implicit function, improving adaptability to variations in scale and resolution. To capture global scene context efficiently, RSMamba [45] introduces state space modeling into RS image classification. Recently, Tri-CoMamba [46] extends Mamba-based modeling to multisource RS classification by fusing spatial, spectral, and modal information. These methods have advanced RS image classification in the single-label setting. However, RS scenes often contain multiple semantic categories within one image, making multilabel classification more challenging, especially under noisy annotations. Accordingly, we investigate robust RS MLC under label noise.

B. RS Multilabel Noise Injection Methods

Existing noise injection methods for RS multilabel image classification mainly follow an instance-independent setting, where label corruption is determined without image content. For instance, uniform noise [18] randomly flips labels with the same probability across images and categories. Partial uniform noise [20] applies uniform flipping only to a randomly selected subset of training images. Distribution-aware noise [19] assigns category-wise flipping probabilities according to the empirical label distribution, making frequent categories more likely to be corrupted. Although these methods are simple and controllable, they poorly capture annotation errors related to image content in RS images [47], [48], [49]. To alleviate this limitation, IDMN [21] introduces the first instance-dependent RS multilabel noise generation method, motivated by the intuition that hard instances are more likely to be mislabeled. However, the noise level in IDMN is not explicitly controlled, since its thresholds are derived solely from the score distribution of present labels. Therefore, we aim to generate instance-dependent noise with controllable levels for more reliable and fair evaluation of noise-robust training methods.

C. RS Multilabel Noise-Robust Methods

In RS multilabel classification, Aksoy et al. [20] propose a collaborative framework where two networks rank training images according to a samplewise loss and exclude images with highly corrupted label sets. To investigate different types of label noise, Burgert et al. [19] categorize multilabel noise into additive, subtractive, and mixed noise. Furthermore, they adapt several single-label robust learning methods to RS multilabel classification, including SAT [50], which refines training targets according to model predictions, ELR [34], which regularizes early learning for each label, and JoCoR [51], which selects labels with small losses via joint regularization. Additionally, recent studies explore robust learning under multilabel noise [52], [53]. Nevertheless, existing methods are primarily designed and evaluated under instance-independent noise, while their behavior under instance-dependent RS multilabel noise remains insufficiently understood.

III. Methodology

In this section, we first formulate the RS MLC task under noisy supervision. Then, we present our DD-IDMN. Finally, we introduce PAIR-Net, a dedicated cross-modal learning framework to address this challenging noise issue.

A. Problem Definition

Let $\mathcal{D} = \{(x_i, \mathbf{y}_i)\}_{i=1}^{N}$ denote a clean RS multilabel training dataset, where x_i is the ith image and $\mathbf{y}_i = [y_{i,1}, \ldots, y_{i,C}] \in \{0,1\}^C$ is its clean label vector over C predefined categories. Here, $y_{i,c} = 1$ indicates that category c is present in image x_i, while $y_{i,c} = 0$ denotes its absence. In practice, only a noisy dataset $\tilde{\mathcal{D}} = \{(x_i, \tilde{\mathbf{y}}_i)\}_{i=1}^{N}$ is available, where $\tilde{\mathbf{y}}_i$ may contain both subtractive and additive noise. Subtractive noise occurs when an existing category is omitted from the annotation, i.e., $y_{i,c} = 1 \rightarrow \tilde{y}_{i,c} = 0$, whereas additive noise occurs when an absent category is falsely annotated as present, i.e., $y_{i,c} = 0 \rightarrow \tilde{y}_{i,c} = 1$. These two noise types may coexist in the same label vector, forming mixed noise with both missed real categories and fabricated false categories. The goal is to train a robust multilabel classification model from the noisy supervision $\tilde{\mathbf{y}}_i$.

B. Dual-Distribution IDMN

To address the thresholding bias in IDMN, we propose the DD-IDMN generation mechanism. Following distribution-aware noise [19], we set $K_\eta = \lfloor N_{\text{pos}} \times \eta \rfloor$ as the target quantity for additive and subtractive noise at a global noise rate $\eta \in (0,1)$, where $N_{\text{pos}} = \sum_{i=1}^{N}\sum_{c=1}^{C} y_{i,c}$ is the number of clean positive labels. Mixed noise therefore contains K_η corruptions of each type, totaling $2K_\eta$ corrupted label entries.

Following IDMN [21], we employ a frozen CLIP model (i.e., ViT-B/32) to compute the prediction score $s_{i,c} = \text{CLIP}(x_i, c)$ for each image and category pair. Unlike IDMN, DD-IDMN explicitly separates the score distributions of present and absent labels. Specifically, all prediction scores are partitioned by the clean labels $y_{i,c}$ into the present-label score set $\mathcal{S}_{\text{pos}}$ and the absent-label score set $\mathcal{S}_{\text{neg}}$

$$\mathcal{S}_{\text{pos}} = \left\{ s_{i,c} \mid y_{i,c} = 1 \right\}, \quad \mathcal{S}_{\text{neg}} = \left\{ s_{i,c} \mid y_{i,c} = 0 \right\} \tag{1}$$

where $i \in \{1, \ldots, N\}$ denotes the training image index and $c \in \{1, \ldots, C\}$ denotes the category index.

Driven by the intuition that hard label instances are more likely to be mislabeled, DD-IDMN selects visually difficult labels from the decoupled score distributions for noise injection. For present labels, lower prediction scores indicate categories that are truly present but visually inconspicuous and difficult to recognize and thus more likely to be missed. Therefore, the lowest scoring labels in the present score set $\mathcal{S}_{\text{pos}}$ are selected as hard positives for subtractive noise. For absent labels, higher prediction scores indicate categories that are more likely to be falsely annotated as present. Accordingly, the highest scoring labels in the absent score set $\mathcal{S}_{\text{neg}}$ are selected as hard negatives for additive noise. Formally, the subtractive threshold ρ_{sub} and additive threshold ρ_{add} are defined as the K_ηth values of the sorted present and absent score sets

$$\rho_{\text{sub}} = \mathcal{S}_{\text{pos}}^{\text{asc}}\left[K_\eta\right], \quad \rho_{\text{add}} = \mathcal{S}_{\text{neg}}^{\text{desc}}\left[K_\eta\right] \tag{2}$$

where $\mathcal{S}_{\text{pos}}^{\text{asc}}$ denotes $\mathcal{S}_{\text{pos}}$ sorted in ascending order and $\mathcal{S}_{\text{neg}}^{\text{desc}}$ denotes $\mathcal{S}_{\text{neg}}$ sorted in descending order. Thus, present labels

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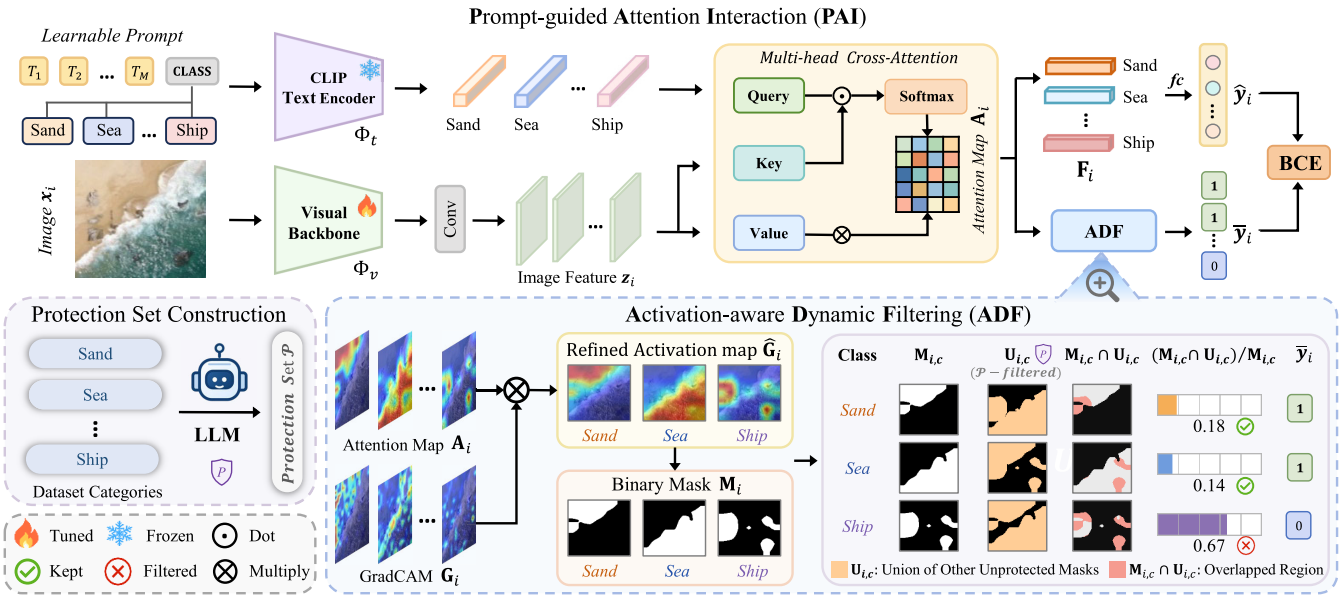


Fig. 2. Illustration of the proposed PAIR-Net. First, the PAI module exploits learnable text prompts and cross-modal attention to enhance class-specific RS representations and generate attention maps for subsequent GradCAM refinement. Then, the ADF module identifies and rectifies potential additive noisy labels with diffuse activations covered by genuine present categories. The Protection Set Construction module protects category pairs with natural spatial inclusion relationships from mistaken filtering. Dashed modules (i.e., protection set construction and ADF) are employed only during training.

with scores below ρ_{sub} are selected as hard positives and flipped from 1 to 0 to generate subtractive noise, while absent labels with scores above ρ_{add} are selected as hard negatives and flipped from 0 to 1 to form additive noise. Since multiple labels may share the boundary score, we further sample labels whose scores equal the corresponding thresholds until the target quantity is exactly matched, yielding the hard positive set $\mathcal{H}_{\text{pos}}$ and hard negative set $\mathcal{H}_{\text{neg}}$ with $|\mathcal{H}_{\text{pos}}| = |\mathcal{H}_{\text{neg}}| = K_\eta$. Formally, the generated noisy label $\tilde{y}_{i,c}$ is defined as

$$\tilde{y}_{i,c} = \begin{cases} 0, & \text{if } (i,c) \in \mathcal{H}_{\text{pos}} \\ 1, & \text{if } (i,c) \in \mathcal{H}_{\text{neg}} \\ y_{i,c}, & \text{otherwise.} \end{cases} \tag{3}$$

By deriving the score thresholds from the decoupled present and absent score distributions, DD-IDMN aligns the numbers of injected subtractive and additive noisy labels with the same target quantity specified by the global noise rate η. This design provides a controllable and balanced synthetic benchmark for evaluating noise-robust multilabel learning methods.

C. Prompt-Guided Activation Interference Rectification Network

To address the instance-dependent noisy supervision simulated by DD-IDMN, we propose the PAIR-Net, a robust multilabel classification framework for RS. As illustrated in Fig. 2, it consists of two components: prompt-guided attention interaction (PAI) and ADF. Specifically, the PAI module integrates learnable text prompts with cross-modal attention to learn class-discriminative RS representations. During training, the ADF module refines GradCAM [54] with cross-modal attention maps and leverages the refined activations to identify unreliable present labels.

1) Prompt-Guided Attention Interaction: To improve class discrimination under noisy supervision, we introduce a PAI module that leverages the semantic priors of a pretrained vision–language model (i.e., CLIP [26]). In PAI, the frozen CLIP text encoder Φ_t produces category-specific text representations serving as semantic anchors, while a trainable RS visual encoder Φ_v extracts spatial feature maps from each input image x_i.

Specifically, for each category c, we construct a learnable prompt $\mathbf{P}_c$ by concatenating M learnable context tokens $\mathbf{T} = [T_1, \ldots, T_M]$ with the category name embedding E_c

$$\mathbf{P}_c = [T_1, \ldots, T_M, E_c]. \tag{4}$$

Then, $\mathbf{P}_c$ is fed into the frozen CLIP text encoder Φ_t to obtain the RS category semantic embedding $\mathbf{t}_c$

$$\mathbf{t}_c = \Phi_t(\mathbf{P}_c) \tag{5}$$

where $\mathbf{t}_c \in \mathbb{R}^d$ denotes the textual representation and d is the embedding dimension. Meanwhile, given an input image x_i, the visual encoder Φ_v extracts a spatial feature map $\mathbf{h}_i$

$$\mathbf{h}_i = \Phi_v(x_i) \tag{6}$$

where $\mathbf{h}_i \in \mathbb{R}^{H \times W \times d_v}$, with H, W, and d_v denoting the spatial height, spatial width, and visual channel dimension, respectively. Since d_v may differ from the text embedding dimension d, we employ a 1×1 convolution to align the visual features with the text embedding space

$$\mathbf{z}_i = \text{Conv}_{1\times 1}(\mathbf{h}_i) \tag{7}$$

where $\mathbf{z}_i \in \mathbb{R}^{H \times W \times d}$ denotes the projected visual feature map. To aggregate spatial features guided by textual category semantics, we reshape $\mathbf{z}_i$ into a token sequence in $\mathbb{R}^{HW \times d}$ and perform cross-modal attention. Specifically, for each category

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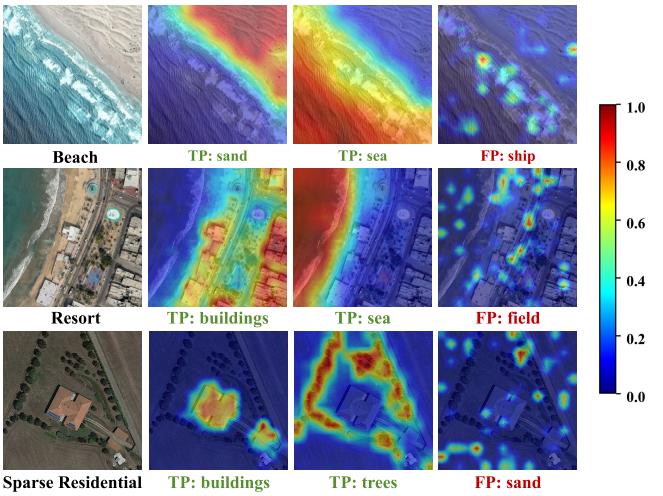


Fig. 3. Comparison of TP and FP activation patterns on representative AID scenes during training. TP activation maps reveal spatially coherent responses aligned with category semantics, whereas FP maps are scattered and largely overlap with TP-activated regions. The color represents the intensity of feature activation, where red indicates high activation levels and blue denotes inactive background.

c, the textual embedding $\mathbf{t}_c$ is projected as the query, while the visual token sequence $\mathbf{z}_i$ is projected as key and value to obtain the cross-modal representation $\mathbf{F}_{i,c} \in \mathbb{R}^d$

$$\mathbf{F}_{i,c} = \text{Softmax}\left(\frac{Q_c K_i^\top}{\sqrt{d}}\right) V_i \tag{8}$$

where $Q_c = \mathbf{t}_c W_q$, $K_i = \mathbf{z}_i W_k$, and $V_i = \mathbf{z}_i W_v$ denote the query, key, and value projections, respectively, with W_q, W_k, and W_v being learnable matrices. Finally, the predicted probability $\hat{y}_{i,c}$ for image x_i is computed as

$$\hat{y}_{i,c} = \sigma\left(\mathbf{w}_c^\top \mathbf{F}_{i,c} + b_c\right) \tag{9}$$

where $\sigma(\cdot)$ is the sigmoid function and $\mathbf{w}_c$ and b_c are the classifier weight and bias for category c, respectively. By injecting category-specific textual priors into visual representation learning, PAI improves RS class discrimination under noisy supervision. Additionally, the averaged multihead cross-modal attention weights are reshaped into a category-specific spatial attention map $\mathbf{A}_{i,c} \in \mathbb{R}^{H \times W}$, which highlights regions relevant to category c. This attention map provides localization guidance for refining the initial GradCAM in the subsequent ADF module.

2) Activation-Aware Dynamic Filtering: Intuitively, in an ideal representation, co-occurring true targets in an RS image should preserve distinct physical boundaries. Accordingly, their activation regions are expected to be spatially coherent with limited overlap. In contrast, since FP labels lack corresponding instances, fitting such labels tends to yield diffuse activations over unrelated regions. As illustrated in Fig. 3, we visualize the activation behavior of true positive (TP) and FP labels during training. The TP activation maps are generally coherent and aligned with visual regions, whereas FP activations are scattered and lack an independent class-specific region. Motivated by this observation, we exploit spatial semantic interference among category activation maps to identify additive noise. Specifically, if the activation region of a positive category is largely contained within activations of other present categories, the corresponding label is more likely to be an FP than a true label supported by independent category-aligned visual content.

To characterize this activation behavior, we compute the GradCAM $\mathbf{G}_{i,c}$ on the visual feature map $\mathbf{z}_i$. For category c and channel k, the importance weight $\alpha_{i,c}^k$ is obtained by globally averaging the gradients of the predicted probability $\hat{y}_{i,c}$ with respect to $z_{i,p,q}^k$ over spatial locations (p, q)

$$\mathbf{G}_{i,c} = \text{ReLU}\left(\sum_k \alpha_{i,c}^k \mathbf{z}_i^k\right), \alpha_{i,c}^k = \frac{1}{HW}\sum_{p=1}^{H}\sum_{q=1}^{W}\frac{\partial \hat{y}_{i,c}}{\partial z_{i,p,q}^k} \tag{10}$$

where $\hat{y}_{i,c}$ denotes the predicted probability and $\mathbf{z}_i^k$ is the kth channel of the projected visual feature map. To obtain more focused activation maps, we utilize the attention map $\mathbf{A}_{i,c}$ to suppress irrelevant spatial responses

$$\hat{\mathbf{G}}_{i,c} = \mathbf{G}_{i,c} \odot \mathbf{A}_{i,c} \tag{11}$$

where $\odot$ denotes element-wise multiplication. The refined activation map $\hat{\mathbf{G}}_{i,c}$ suppresses nontarget responses and provides reliable activation localization for our dynamic noise filtering.

Then, the refined activation map $\hat{\mathbf{G}}_{i,c}$ is binarized by Otsu's thresholding [55], which automatically determines the threshold $\theta_{i,c}$ for each category-specific activation map. Formally, the binary mask $\mathbf{M}_{i,c}$ at each spatial location (p, q) follows:

$$\mathbf{M}_{i,c}(p,q) = \begin{cases} 1, & \hat{\mathbf{G}}_{i,c}(p,q) \geq \theta_{i,c} \\ 0, & \text{otherwise.} \end{cases} \tag{12}$$

To quantify cross-class semantic interference among positive labels, we propose the **SCS**, which measures how much the activation of target category c is covered by other present categories in image x_i. A high **SCS** indicates target activation largely contained within other category activations, suggesting weak class discrimination and a higher likelihood of additive noise. Conversely, a low **SCS** indicates distinct target activation and category focusing, suggesting that the label is more likely to be a TP.

Since geographical entities in RS imagery naturally exhibit spatial inclusion relationships, directly treating high **SCS** values as noisy labels may suppress hard positives. For example, airplanes often appear on pavement, and ships are located within water. Such pairs may produce large activation overlaps even when both labels are clean. To prevent valid spatial overlap from being treated as noisy interference, we construct a dataset-specific semantic protection set $\mathcal{P}$ using an LLM (i.e., ChatGPT powered by GPT-5.5 [29]). As described in Fig. 4, the construction pipeline combines a prompt template with human validation to verify inclusion relationships and category names. Although physical inclusion relationships are inherently directional, we define $\mathcal{P}$ as a symmetric unordered set since our protection mechanism removes valid activation overlap between category pairs from the **SCS** calculation. Accordingly, for target present class c in image x_i, we define

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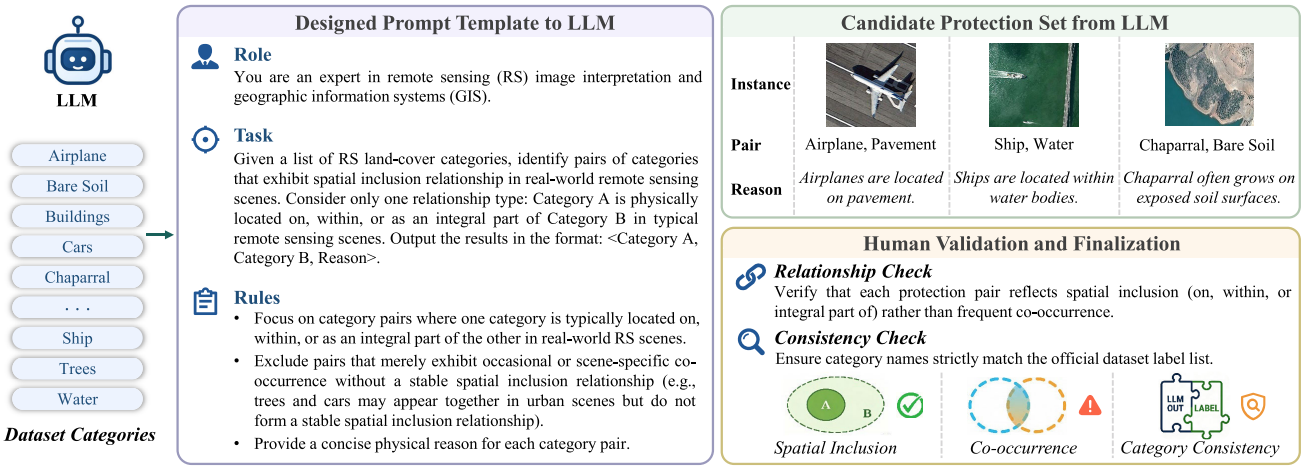


Fig. 4. Construction pipeline of the RS-specific semantic protection set $\mathcal{P}$. Specifically, we utilize an LLM (i.e., ChatGPT powered by GPT-5.5 [29]) to identify category pairs with stable physical inclusion relationships from the predefined dataset categories. The generated candidates are then manually verified through a *Relationship Check*, which distinguishes genuine physical inclusion (e.g., ship–water) from frequent co-occurrence without a stable spatial inclusion relation (e.g., sand–sea), and a *Consistency Check*, which maps all category names to the official dataset labels.

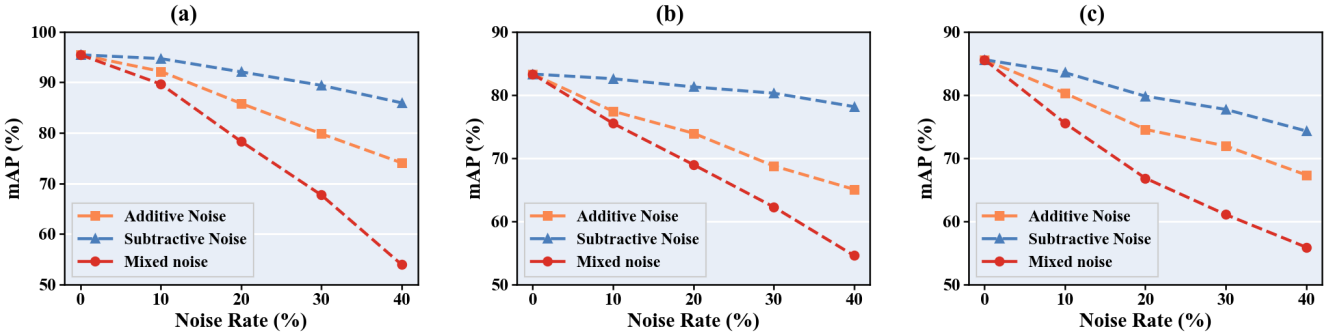


Fig. 5. Classification performance of different noise types under DD-IDMN across varying noise rates on (a) UCMerced, (b) AID, and (c) MLRSNet datasets. Under the DD-IDMN noise setting, additive noise leads to more severe performance degradation than subtractive noise across all evaluated datasets. Mixed noise yields the lowest performance, containing both K_η additive and K_η subtractive corruptions at noise rate η.

the epoch-dependent unprotected set $\Omega_{i,c}^{(t)}$ by excluding protected pairs with c from current present labels $\bar{y}_{i,k}^{(t)}$

$$\Omega_{i,c}^{(t)} = \left\{ k \mid \bar{y}_{i,k}^{(t)} = 1,\ k \neq c,\ \{c,k\} \notin \mathcal{P} \right\} \tag{13}$$

where $\bar{y}_{i,k}^{(t)}$ denotes the current label state at epoch t, with $\bar{y}_{i,k}^{(t)} = 1$ indicating that category k is annotated as present for image x_i. Formally, at training epoch t, let $\mathbf{M}_{i,c}^{(t)}$ denote the binary activation mask of the target positive category c, and let $\mathbf{U}_{i,c}^{(t)}$ denote the union mask of the corresponding unprotected present categories in $\Omega_{i,c}^{(t)}$ under the current label state. **SCS** is defined as the fraction of target mask pixels in $\mathbf{M}_{i,c}^{(t)}$ that overlap with the union mask $\mathbf{U}_{i,c}^{(t)}$

$$\mathbf{SCS}_{i,c}^{(t)} = \frac{\left|\mathbf{M}_{i,c}^{(t)} \cap \mathbf{U}_{i,c}^{(t)}\right|}{\left|\mathbf{M}_{i,c}^{(t)}\right|}, \quad \mathbf{U}_{i,c}^{(t)} = \bigcup_{k \in \Omega_{i,c}^{(t)}} \mathbf{M}_{i,k}^{(t)} \tag{14}$$

where $\mathbf{SCS}_{i,c}^{(t)} = 0$ if $|\mathbf{M}_{i,c}^{(t)}| = 0$ or $\Omega_{i,c}^{(t)} = \varnothing$. Finally, we rectify potential additive noisy labels with a predefined filtering threshold τ, yielding the updated label $\bar{y}_{i,c}^{(t+1)}$

$$\bar{y}_{i,c}^{(t+1)} = \begin{cases} 0, & \bar{y}_{i,c}^{(t)} = 1 \text{ and } \mathbf{SCS}_{i,c}^{(t)} > \tau \\ \bar{y}_{i,c}^{(t)}, & \text{otherwise.} \end{cases} \tag{15}$$

After T_w warm-up epochs, the label state is initialized with the observed noisy labels as $\bar{y}_{i,c}^{(T_w+1)} = \tilde{y}_{i,c}$. During the filtering stage at epoch t, we update the rectified label vector $\bar{\mathbf{y}}_i^{(t+1)}$. Only positive labels identified as unreliable are flipped from 1 to 0, while all other labels remain unchanged. Thus, the proposed ADF module progressively suppresses additive noisy labels during training, mitigating the severe degradation caused by additive noise, as empirically demonstrated in Fig. 5.

3) Optimization Objective: At each training epoch t, PAIR-Net is supervised by the current label vector $\bar{\mathbf{y}}_i^{(t)}$. Given a mini-batch of B images and C predefined categories, the binary cross-entropy (BCE) objective is formulated as

$$\mathcal{L}^{(t)} = -\frac{1}{B}\sum_{i=1}^{B}\sum_{c=1}^{C}\left[\bar{y}_{i,c}^{(t)} \log \hat{y}_{i,c}^{(t)} + \left(1 - \bar{y}_{i,c}^{(t)}\right) \log\left(1 - \hat{y}_{i,c}^{(t)}\right)\right]. \tag{16}$$

To present an intuitive understanding of our PAIR-Net, the overall procedure is summarized in Algorithm 1.

IV. Experiment

In this section, we present the experimental settings, including datasets, evaluation metrics, implementation details, and compared methods. We then compare DD-IDMN with

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Algorithm 1 PAIR-Net

Require: Training dataset $\tilde{\mathcal{D}} = \{(x_i, \tilde{\mathbf{y}}_i)\}_{i=1}^{N}$, semantic protection set $\mathcal{P}$, network parameters Θ, warm-up epochs T_w, filtering epochs T_f, total epochs T_{total}, batch size B.

Initialize $\bar{\mathbf{y}}_i^{(T_w+1)} \leftarrow \tilde{\mathbf{y}}_i$
for $t = 1, 2, \ldots, T_{total}$ **do**
  # Warm-up Training
  if $t \leq T_w$ **then**
    for *iteration* $= 1, 2, \ldots$ **do**
      Fetch $\mathcal{B} = \{(x_i, \tilde{\mathbf{y}}_i)\}_{i=1}^{B}$ from $\tilde{\mathcal{D}}$;
      Calculate $\mathcal{L}^{(t)} = \frac{1}{|\mathcal{B}|} \sum \ell^{(t)}(x_i, \tilde{y}_i)$; ▷ Eq. (16)
      Update Θ by optimizing $\mathcal{L}^{(t)}$.
    end for
  end if
  # Robust Training
  if $T_w < t \leq T_{total}$ **then**
    for *iteration* $= 1, 2, \ldots$ **do**
      Fetch $\mathcal{B} = \{(x_i, \bar{\mathbf{y}}_i^{(t)})\}_{i=1}^{B}$;
      # Prompt-guided Attention Interaction
      Obtain cross-modal feature $\mathbf{F}_i$ ▷ Eq. (8)
      if $t \leq T_w + T_f$ **then**
        # Activation-aware Dynamic Filtering
        Obtain cross-modal attention map $\mathbf{A}_i$;
        Refine GradCAM $\mathbf{G}_i$ with $\mathbf{A}_i$; ▷ Eq. (11)
        Generate binary mask $\mathbf{M}_i$; ▷ Eq. (12)
        Calculate $\mathbf{SCS}_i$; ▷ Eq. (14)
        Obtain rectified labels $\bar{\mathbf{y}}_i^{(t+1)}$; ▷ Eq. (15)
      else
        Retain current labels: $\bar{\mathbf{y}}_i^{(t+1)} \leftarrow \bar{\mathbf{y}}_i^{(t)}$;
      end if
      Calculate $\mathcal{L}^{(t)} = \frac{1}{|\mathcal{B}|} \sum \ell^{(t)}(x_i, \bar{\mathbf{y}}_i^{(t)})$; ▷ Eq. (16)
      Update Θ by optimizing $\mathcal{L}^{(t)}$.
    end for
  end if
end for

different noise injection strategies and evaluate PAIR-Net against noise-robust learning methods. Finally, comprehensive ablation studies are conducted to systematically analyze the effectiveness and characteristics of the proposed approach.

A. Datasets

We conduct experiments on three multilabel RS image datasets: UCMerced [56], AID [57], and MLRSNet [58]. UCMerced and AID were originally designed for single-label scene classification, and we adopt their public multilabel annotations from previous studies [59], [60]. MLRSNet is a large-scale RS dataset with native multilabel annotations. These datasets vary in image scale and scene complexity, enabling a comprehensive evaluation across multilabel scenarios.

1) *UCMerced [56]:* The original dataset contains 2100 aerial images from the United States Geological Survey National Map, each with 256×256 pixels at 0.3-m resolution. We adopt the manually interpreted 17-category multilabel annotations from [59]. It is split into 1680 training and 420 test images.
2) *AID [57]:* It contains 10 000 Google Earth images of 600×600 pixels at 0.5–8-m resolution. Following previous studies, we use the AID multilabel subset from [60], where 3000 images are annotated with 17 categories. This multilabel subset is split into 2400 training and 600 test images.
3) *MLRSNet [58]:* It is a large-scale multilabel RS image dataset collected from Google Earth. It contains 109 161 images of 256×256 pixels, with pixel resolutions ranging from 0.1 to 10 m. Each image is annotated with multiple labels from 60 categories, making MLRSNet more challenging due to its larger data scale and richer category space. The dataset is split into 87 328 training and 21 833 test images.

B. Evaluation Metrics

Following IDMN [21], we report macro-averaged mean average precision (mAP) and Hamming loss (Ham.) metrics. All results are reported as mean $\pm$ std over three random runs under identical settings, with seeds in $(0, 1, 2)$.

C. Implementation Details

Following IDMN [21], ResNet-50 initialized with ImageNet weights is adopted as the visual backbone Φ_v for all comparisons. The network is optimized with Adam [61] for 50 epochs, with an initial learning rate of 1×10^{-4}. All models are trained on a single NVIDIA RTX 3090 GPU with a batch size of 16 across all experiments. Standard data augmentation consists of random horizontal flipping with a probability of 0.5. DD-IDMN uses frozen CLIP ViT-B/32 for noise generation, while PAIR-Net employs a ViT-L/14 text encoder with an embedding dimension of 768 during training.

D. Compared Methods

We compare DD-IDMN with various noise strategies and evaluate PAIR-Net against representative robust methods.

1) *Different Noise Injection Strategies:* We compare DD-IDMN with four multilabel noise strategies: uniform noise [18], partial uniform noise [20], distribution-aware noise [19], and IDMN [21]. Given a noise rate η, uniform noise randomly samples label entries, whereas partial uniform noise restricts sampling to selected training images. Distribution-aware noise samples labels class-wise according to positive-label counts. IDMN uses the CLIP score distribution of present labels to select low-score positives and high-score negatives. For fair comparison, we adjust the instance-independent methods to match DD-IDMN in the target quantities of additive and subtractive corruptions at each noise rate while retaining their sampling strategies. Since IDMN does not explicitly control its noise quantity, we report the actual number of corruptions.
2) *Noise-Robust Learning Methods:* Following IDMN [21], we compare PAIR-Net with standard BCE, label correlation regularization (LCR [18]), robust collaborative

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TABLE I

COMPARISON OF MULTILABEL CLASSIFICATION PERFORMANCE ACROSS SYNTHETIC MULTILABEL NOISE GENERATION METHODS AND NOISE LEVELS USING BCE AS THE BASELINE METHOD. MIX-η DENOTES THE TARGET NOISE QUANTITY K_η FOR BOTH ADDITIVE AND SUBTRACTIVE CORRUPTION. RESULTS ARE REPORTED AS THE MEAN $\pm$ STD OVER THREE RANDOM RUNS WITH SEEDS IN $(0, 1, 2)$

Datasets	Noise Model	Mix-10%		Mix-20%		Mix-30%		Mix-40%	
		mAP↑ (%)	Ham.↓ (%)	mAP↑ (%)	Ham.↓ (%)	mAP↑ (%)	Ham.↓ (%)	mAP↑ (%)	Ham.↓ (%)
UCM	Uniform	91.97±0.80	6.68±0.19	88.44±0.37	7.96±0.03	82.57±1.61	9.80±0.22	75.05±0.11	11.66±0.25
	Partial uniform	91.93±0.97	5.82±0.25	90.38±0.85	6.37±0.01	87.17±1.29	7.57±0.50	85.26±0.58	8.54±0.19
	Distribution-aware	92.35±0.32	7.02±0.30	89.14±1.01	9.21±0.41	85.01±0.88	11.74±0.43	77.68±0.11	15.44±0.30
	IDMN	90.80±0.38	7.61±0.33	83.66±0.79	10.17±0.13	70.60±0.88	13.64±0.36	58.74±1.33	18.06±0.49
	DD-IDMN (ours)	89.65±0.15	8.10±0.09	78.31±0.98	11.46±0.12	67.71±0.73	14.65±0.19	53.96±0.70	17.65±0.18
AID	Uniform	80.19±0.28	6.81±0.10	75.10±0.30	8.54±0.13	66.77±0.96	11.60±0.27	55.91±0.71	16.36±0.52
	Partial uniform	80.58±0.57	6.19±0.04	77.74±0.93	6.45±0.16	75.08±0.71	7.32±0.11	72.39±0.52	8.25±0.46
	Distribution-aware	80.43±0.61	8.23±0.37	73.63±0.38	11.79±0.26	65.03±0.20	16.08±0.47	57.88±0.86	20.59±0.16
	IDMN	76.34±0.49	9.30±0.04	69.88±0.07	13.34±0.05	63.90±0.36	17.96±0.17	55.94±0.26	23.53±0.06
	DD-IDMN (ours)	75.85±0.61	9.41±0.15	68.96±0.09	14.44±0.28	62.29±0.90	19.48±0.14	54.74±0.43	24.37±0.09
MLRSNet	Uniform	80.31±0.42	4.22±0.04	76.91±0.35	4.88±0.05	72.00±0.36	5.77±0.03	65.26±0.25	6.71±0.01
	Partial uniform	77.23±0.73	3.96±0.03	72.55±0.09	4.48±0.05	68.79±0.36	5.46±0.03	65.65±0.38	7.29±0.16
	Distribution-aware	80.80±0.47	4.44±0.07	78.08±0.04	5.37±0.03	73.92±0.14	6.50±0.06	68.09±0.40	7.89±0.02
	IDMN	73.73±0.07	4.88±0.03	63.29±0.12	6.83±0.04	54.33±0.13	9.60±0.01	45.56±0.12	13.60±0.10
	DD-IDMN (ours)	75.53±0.20	4.68±0.02	66.84±0.19	5.98±0.01	61.09±0.22	7.19±0.02	55.92±0.23	8.54±0.05

multilabel learning (RCML [20]), self-adaptive training for MLC (SAT-ML [19]), early learning regularization (ELR-ML [19]), and joint training with co-regularization (JoCoR-ML [19]).

E. Effect of Different Noise Types Under DD-IDMN

We investigate the effects of different noise types under DD-IDMN. As illustrated in Fig. 5, mixed noise consistently yields the lowest mAP across all three datasets. It contains both additive noise and subtractive noise, making it the most complex and challenging noise setting.

Additionally, Fig. 5 shows a clear gap between additive and subtractive noise under the DD-IDMN noise setting, with additive noise yielding greater performance degradation than subtractive noise. In DD-IDMN, additive noise is produced by flipping visually confusing hard negative labels to positive ones. Once these labels are assigned as positive labels, the model is explicitly encouraged to associate image regions with incorrect categories, which directly introduces misleading semantic supervision during training. In contrast, subtractive noise removes hard positive labels that are difficult to recognize. Although it weakens the supervision for some true categories, it mainly leads to insufficient learning rather than forcing the model to fit incorrect category signals during training. These observations motivate PAIR-Net to focus its explicit noise filtering mechanism on the more destructive additive noise, thereby mitigating the dominant effect of label noise under the DD-IDMN noise setting.

F. Comparison With Different Noise Injection Strategies

We compare DD-IDMN with different RS multilabel noise generation methods using the BCE baseline method under the same experimental settings. As shown in Table I, DD-IDMN yields lower mAP and higher Hamming loss than conventional instance-independent methods, including uniform, partial uniform, and distribution-aware noise models, across different noise levels and datasets. These results indicate that our DD-IDMN provides a more challenging synthetic benchmark for evaluating model robustness.

More importantly, DD-IDMN provides more precise noise control than IDMN. As demonstrated in Fig. 6, the actual number of noisy labels generated by IDMN deviates from the target quantity, especially on the more challenging MLRSNet dataset. This observation is consistent with our analysis that the number of noisy labels generated by IDMN may deviate from the intended target, resulting in an unbalanced benchmark. In contrast, DD-IDMN explicitly decouples the generation of additive and subtractive noise, allowing it to match the additive and subtractive corruption quantities across datasets. This design enables a fairer and more systematic evaluation of model robustness under different types of multilabel noise.

G. Comparison With Different Noise-Robust Learning Methods

To evaluate PAIR-Net under DD-IDMN, we compare it with representative noise-robust multilabel learning methods on UCMerced [56], AID [57], and MLRSNet [58]. As demonstrated in Table II, LCR and ELR-ML provide limited gains over BCE, suggesting that label correlation and early learning regularization are insufficient for the challenging mixed noise generated by DD-IDMN. Additionally, RCML and JoCoR-ML obtain lower Hamming loss in some cases, especially on UCMerced and AID, indicating that collaborative training can help reduce label-wise prediction errors. However, these gains are not consistently reflected in mAP, showing limited overall robustness to challenging instance-dependent noise.

In contrast, SAT-ML achieves more competitive performance, indicating that self-adaptive target refinement is

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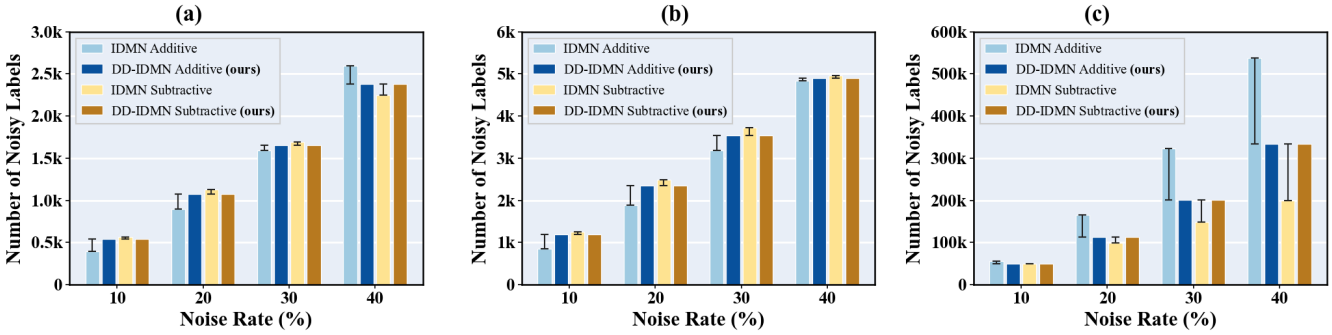


Fig. 6. Comparison of actual noisy label quantities between IDMN and our DD-IDMN across different datasets and noise rates. Following [19], the target quantity is defined as the number of present labels multiplied by the noise rate η. The vertical black bars denote deviations from the target quantity. IDMN cannot precisely control the number of additive and subtractive noisy labels, especially on MLRSNet. In contrast, DD-IDMN accurately matches the target quantity for both noise types, enabling a more controllable and balanced synthetic evaluation benchmark. (a) UCMerced. (b) AID. (c) MLRSNet.

TABLE II

COMPARISON WITH NOISE-ROBUST METHODS UNDER MIXED DD-IDMN NOISE ON UCMERCED, AID, AND MLRSNET. MIX-η DENOTES THE TARGET NOISE QUANTITY K_η FOR BOTH ADDITIVE AND SUBTRACTIVE CORRUPTION. RESULTS ARE MEAN ± STD OVER THREE RUNS WITH SEEDS (0, 1, 2)

Datasets	Algorithms	Mix-10%		Mix-20%		Mix-30%		Mix-40%	
		mAP↑ (%)	Ham.↓ (%)	mAP↑ (%)	Ham.↓ (%)	mAP↑ (%)	Ham.↓ (%)	mAP↑ (%)	Ham.↓ (%)
UCM	BCE	89.65±0.15	8.10±0.09	78.31±0.98	11.46±0.12	67.71±0.73	14.65±0.19	53.96±0.70	17.65±0.18
	LCR	90.64±0.45	7.90±0.14	77.51±0.67	11.37±0.25	67.99±0.59	14.30±0.09	54.93±0.74	17.50±0.09
	RCML	90.60±0.22	**7.27±0.13**	78.73±0.83	10.80±0.23	68.89±1.50	13.67±0.26	56.09±1.35	17.32±0.13
	SAT-ML	92.56±0.13	7.61±0.10	83.75±0.32	10.69±0.16	73.97±0.09	13.12±0.06	62.40±0.33	16.12±0.09
	ELR-ML	89.90±0.55	8.00±0.14	77.92±1.34	11.33±0.14	67.73±0.45	13.12±0.06	54.24±0.45	17.51±0.16
	JoCoR-ML	89.38±1.50	7.39±0.33	78.58±0.55	10.75±0.17	70.24±0.75	12.95±0.20	59.13±0.53	16.40±0.40
	PAIR-Net (ours)	**93.34±0.70**	7.36±0.15	**87.42±0.48**	**10.16±0.09**	**81.27±0.29**	**11.96±0.29**	**75.53±0.60**	**14.41±0.06**
AID	BCE	75.85±0.61	9.41±0.15	68.96±0.09	14.44±0.28	62.29±0.90	19.48±0.14	54.74±0.43	24.37±0.09
	LCR	73.24±0.21	9.39±0.04	66.88±0.84	13.96±0.14	59.97±0.42	18.96±0.16	53.85±0.14	23.76±0.04
	RCML	76.57±0.41	9.26±0.11	70.51±0.26	14.08±0.11	63.16±0.18	19.54±0.30	56.27±0.36	24.25±0.11
	SAT-ML	77.23±0.45	**8.83±0.08**	70.80±0.24	13.55±0.11	64.44±0.36	18.46±0.12	57.03±0.25	23.56±0.18
	ELR-ML	75.75±0.65	9.43±0.14	68.94±0.34	14.49±0.07	61.63±0.85	19.49±0.18	55.11±0.39	24.38±0.04
	JoCoR-ML	75.23±0.46	8.98±0.17	69.72±0.43	13.48±0.31	63.13±0.44	18.39±0.23	56.17±0.14	23.06±0.20
	PAIR-Net (ours)	**78.43±0.67**	8.85±0.13	**75.18±0.82**	**12.46±0.12**	**68.72±0.54**	**15.90±0.25**	**63.91±0.44**	**19.45±0.07**
MLRSNet	BCE	75.53±0.20	4.68±0.02	66.84±0.19	5.98±0.01	61.09±0.22	7.19±0.02	55.92±0.23	8.54±0.05
	LCR	75.43±0.37	4.70±0.03	66.82±0.21	5.97±0.01	61.03±0.06	7.20±0.03	55.61±0.22	8.49±0.04
	RCML	74.76±0.56	4.87±0.02	65.74±0.38	5.96±0.02	62.39±0.22	7.00±0.01	57.62±0.01	8.18±0.02
	SAT-ML	80.79±0.30	4.31±0.02	72.08±0.19	5.50±0.01	65.84±0.03	6.68±0.04	60.33±0.11	7.84±0.01
	ELR-ML	75.54±0.05	4.69±0.03	66.96±0.17	5.97±0.01	61.30±0.27	7.19±0.03	55.59±0.11	8.51±0.05
	JoCoR-ML	78.56±0.23	4.47±0.02	69.28±0.32	5.48±0.01	60.66±0.01	6.55±0.00	53.96±0.07	7.65±0.02
	PAIR-Net (ours)	**82.70±0.30**	**4.10±0.01**	**75.93±0.94**	**5.16±0.05**	**68.61±0.25**	**6.26±0.01**	**62.91±0.52**	**7.46±0.01**

beneficial for DD-IDMN. By updating training targets with an exponential moving average of model predictions, SAT-ML reduces the direct influence of unreliable labels. Nevertheless, PAIR-Net achieves the best mAP on all evaluated datasets and the best or second-best Hamming loss in most cases, especially at higher noise rates (i.e., 30% and 40%). The results demonstrate that PAIR-Net effectively suppresses noisy labels and maintains reliable learning under mixed DD-IDMN.

We further evaluate all methods under additive and subtractive noise generated by DD-IDMN. As shown in Table III, PAIR-Net consistently achieves the best mAP under additive noise, confirming its ability to suppress destructive added labels. This result is consistent with the design motivation of PAIR-Net, which explicitly filters unreliable positive supervision caused by additive noise. Under subtractive noise, PAIR-Net also remains competitive, as our PAI strengthens category-specific representations through textual semantic guidance and helps infer visually present yet unlabeled categories. Overall, the results under additive, subtractive, and mixed noise demonstrate the robustness of PAIR-Net under DD-IDMN.

H. Generalization Across Diverse Noise Generation Strategies

To further validate the generalization capability of PAIR-Net beyond DD-IDMN, we conduct experiments on UCMerced

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TABLE III

COMPARISON WITH DIFFERENT ROBUST LEARNING METHODS UNDER ADDITIVE AND SUBTRACTIVE DD-IDMN NOISE ON THE UCMERCED (UCM), AID, AND MLRSNET DATASETS. RESULTS ARE REPORTED AS THE MEAN ± STD OVER THREE RANDOM RUNS WITH SEEDS IN (0, 1, 2)

Datasets	Algorithms	Additive-20%		Additive-40%		Subtractive-20%		Subtractive-40%	
		mAP↑ (%)	Ham.↓ (%)	mAP↑ (%)	Ham.↓ (%)	mAP↑ (%)	Ham.↓ (%)	mAP↑ (%)	Ham.↓ (%)
UCM	BCE	85.82±0.38	8.34±0.07	74.07±0.83	12.53±0.21	91.56±0.03	7.61±0.09	85.44±0.32	10.33±0.30
	LCR	85.39±0.66	8.32±0.09	75.78±0.52	12.68±0.04	91.87±0.20	7.39±0.02	85.99±0.56	**9.80±0.03**
	RCML	85.71±0.93	8.22±0.18	77.11±1.07	11.80±0.22	91.06±0.59	7.89±0.21	87.36±0.61	10.47±0.29
	SAT-ML	90.06±0.26	7.65±0.08	81.91±0.16	11.08±0.04	92.87±0.05	7.63±0.05	88.41±0.06	10.44±0.03
	ELR-ML	84.83±0.44	8.67±0.01	74.84±0.04	13.05±0.04	91.85±0.15	7.45±0.08	86.26±0.22	9.81±0.05
	JoCoR-ML	87.09±0.56	**7.38±0.10**	77.47±0.76	10.33±0.28	89.11±0.02	8.44±0.22	80.51±0.61	11.02±0.08
	PAIR-Net (ours)	**91.27±0.32**	7.64±0.29	**87.37±0.50**	**9.67±0.89**	**93.23±0.29**	**7.01±0.58**	**89.87±0.14**	10.09±0.01
AID	BCE	73.95±0.45	9.83±0.03	65.07±0.28	14.37±0.10	81.27±0.33	9.67±0.40	78.15±0.20	15.03±0.19
	LCR	73.19±0.22	9.64±0.05	65.17±0.24	14.24±0.08	81.62±0.07	9.91±0.14	77.85±0.10	14.98±0.06
	RCML	75.51±0.65	9.38±0.04	65.99±0.54	14.41±0.08	81.11±0.19	10.07±0.08	77.08±0.25	14.84±0.24
	SAT-ML	74.84±0.16	8.59±0.06	68.15±0.20	13.25±0.14	82.31±0.09	10.29±0.44	78.75±0.07	15.44±0.09
	ELR-ML	73.91±0.30	9.85±0.10	65.05±0.09	14.46±0.10	81.49±0.29	10.47±0.09	77.75±0.07	15.11±0.02
	JoCoR-ML	75.08±0.65	8.95±0.02	66.19±0.34	13.40±0.19	80.89±0.09	10.08±0.07	76.90±0.17	15.38±0.08
	PAIR-Net (ours)	**77.61±0.28**	**8.21±0.06**	**73.54±0.57**	**10.63±0.58**	**83.82±0.25**	**9.44±0.18**	**79.43±0.31**	**14.67±0.43**
MLRSNet	BCE	75.78±0.24	4.73±0.03	68.87±0.43	6.13±0.10	79.69±0.11	4.75±0.01	73.21±0.17	6.06±0.04
	LCR	74.88±0.25	4.60±0.10	68.30±0.14	6.00±0.01	79.84±0.02	4.73±0.01	73.16±0.05	6.12±0.02
	RCML	74.30±0.03	4.80±0.02	68.68±0.12	5.89±0.02	76.58±0.28	4.93±0.01	71.14±0.01	6.22±0.01
	SAT-ML	81.43±0.10	4.06±0.01	75.89±0.09	5.01±0.02	85.40±0.02	4.62±0.02	79.92±0.02	6.10±0.01
	ELR-ML	75.33±0.30	4.69±0.01	68.14±0.07	5.99±0.03	79.72±0.04	4.73±0.01	73.13±0.02	6.12±0.01
	JoCoR-ML	78.19±0.55	4.14±0.03	71.32±0.38	4.67±0.07	75.70±0.30	5.26±0.03	64.62±0.45	7.03±0.04
	PAIR-Net (ours)	**82.45±0.22**	**3.77±0.01**	**78.79±0.09**	**4.60±0.01**	**87.91±0.35**	**4.07±0.01**	**83.05±0.21**	**5.30±0.01**

and AID datasets under diverse mixed noise scenarios. The evaluated noise generation methods include instance-independent noise models (i.e., uniform, partial uniform, and distribution-aware noise) and the instance-dependent IDMN method, thereby covering corruption guided by both random sampling and image content. For each strategy, we compare PAIR-Net with representative noise-robust multilabel learning methods under different mixed noise rates. As demonstrated in Table IV, PAIR-Net maintains robust performance across different noise generation strategies. Under instance-independent noise settings, PAIR-Net achieves competitive or superior performance in most cases, indicating that the proposed prompt-guided RS feature representation and activation-aware filtering strategy can also reduce the influence of randomly corrupted label entries. Furthermore, under IDMN, PAIR-Net obtains consistent gains on both UCMerced and AID, especially at higher noise rates such as Mix-40%. These results indicate that PAIR-Net is not specialized to our DD-IDMN. Instead, it provides a general robust learning framework for RS multilabel classification under diverse noisy supervision patterns.

I. Ablation Study

In all main experiments, PAIR-Net adopts the same configuration across datasets and noise rates. To analyze each component and configuration, we conduct ablation studies on UCMerced and AID under mixed noise settings. Specifically, we evaluate PAIR-Net components and parameters, including the text encoder, learnable context tokens, attention heads, filtering threshold, warm-up duration, and filtering duration.

1) *Impact of Different Components:* We evaluate each component in PAIR-Net under mixed noise rates. As detailed in Table V, introducing the PAI module consistently improves the performance over BCE. This result indicates that cross-modal semantic priors enhance RS class discrimination under noisy supervision. Although GradCAM-guided filtering brings further gains by identifying unreliable positive labels, replacing it with refined activation maps $\hat{\mathbf{G}}_{i,c}$ in our dynamic filtering ADF module improves performance. This improvement demonstrates that refined activation maps provide reliable spatial evidence for filtering additive noise. Furthermore, we isolate the effect of ADF without the textual prior from PAI, denoted as PAIR-Net w/o PAI. This variant still improves over BCE on both datasets, showing that ADF can enhance noise robustness without cross-modal interaction. Overall, PAIR-Net achieves the best performance, indicating that PAI and ADF provide complementary benefits under noisy multilabel supervision.
2) *Impact of the Number of Learnable Context Tokens:* We evaluate the effect of the prompt context length M on PAIR-Net. As shown in Table VI, learnable prompts generally outperform the fixed handcrafted prompt (i.e., "an aerial picture of [CLASS]" with $M = 0$). This result indicates that learnable context tokens improve the context modeling capacity of text prompts, helping the model capture domain-specific semantics for RS categories. However, increasing M to 20 does not bring consistent further improvement. This phenomenon can be attributed to the fact that excessive context

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TABLE IV

GENERALIZATION ACROSS DIVERSE NOISE GENERATION STRATEGIES ON UCMERCED (UCM) AND AID. MIX-η DENOTES THE TARGET NOISE QUANTITY K_η FOR BOTH ADDITIVE AND SUBTRACTIVE CORRUPTION. RESULTS ARE MEAN ± STD OVER THREE RUNS WITH SEEDS (0, 1, 2)

Datasets	Noise Model	Algorithm	Mix-10%		Mix-20%		Mix-30%		Mix-40%	
			mAP↑ (%)	Ham.↓ (%)	mAP↑ (%)	Ham.↓ (%)	mAP↑ (%)	Ham.↓ (%)	mAP↑ (%)	Ham.↓ (%)
UCM	Uniform	BCE	91.97±0.80	6.68±0.19	88.44±0.37	7.96±0.03	82.57±1.61	9.80±0.22	75.05±0.11	11.66±0.25
		RCML	91.70±0.76	6.89±0.47	89.35±0.96	7.92±0.47	84.54±1.13	9.80±0.25	76.62±2.12	11.39±0.71
		SAT-ML	93.15±0.36	6.75±0.25	91.31±0.64	7.94±0.23	88.47±1.17	10.28±0.11	83.73±0.35	12.47±0.51
		JoCoR-ML	93.49±0.28	**6.28±0.27**	91.02±0.72	**7.83±0.47**	87.66±0.75	9.58±0.24	80.93±0.86	11.62±0.32
		PAIR-Net (ours)	**93.81±0.64**	6.44±0.80	**91.96±0.38**	7.88±0.49	**88.97±0.94**	**9.35±0.29**	**84.42±0.61**	**10.24±0.16**
	Partial uniform	BCE	91.93±0.97	5.82±0.25	90.38±0.85	6.37±0.01	87.17±1.29	7.57±0.50	85.26±0.58	8.54±0.19
		RCML	93.41±0.43	5.57±0.21	92.09±0.29	5.93±0.16	88.05±1.48	7.24±0.47	85.48±1.62	8.59±0.39
		SAT-ML	92.24±0.88	6.23±0.31	90.74±0.61	6.83±0.14	88.49±0.54	7.13±0.26	85.26±0.31	**7.53±0.13**
		JoCoR-ML	93.13±0.72	5.64±0.22	91.41±0.80	6.23±0.29	87.69±0.58	7.63±0.41	83.08±1.04	9.56±0.55
		PAIR-Net (ours)	**94.09±0.28**	**5.33±0.78**	**92.30±0.51**	**5.32±0.33**	**89.08±0.76**	**6.91±0.10**	**87.89±1.43**	8.40±0.43
	Distribution-aware	BCE	92.35±0.32	7.02±0.30	89.14±1.01	9.21±0.41	85.01±0.88	11.74±0.43	77.68±0.11	15.44±0.30
		RCML	91.90±0.35	6.93±0.29	89.29±0.40	9.15±0.52	85.24±0.81	11.72±0.39	78.92±0.68	15.18±0.25
		SAT-ML	93.16±0.20	6.61±0.28	92.03±0.46	8.22±0.09	88.12±0.55	10.27±0.37	83.70±0.55	14.59±0.47
		JoCoR-ML	93.34±0.36	6.09±0.29	91.42±0.45	8.12±0.37	87.48±0.72	10.87±0.42	80.11±0.50	15.55±0.60
		PAIR-Net (ours)	**93.71±0.75**	**6.29±0.54**	**92.80±0.40**	**9.09±0.24**	**88.92±0.63**	**10.16±0.69**	**84.86±0.27**	**12.11±0.71**
	IDMN	BCE	90.80±0.38	7.61±0.33	83.66±0.79	10.17±0.13	70.60±0.88	13.64±0.36	58.74±1.33	18.06±0.49
		RCML	89.99±0.27	7.80±0.25	85.20±0.16	10.36±0.33	72.39±0.43	12.96±0.18	60.94±0.13	17.71±0.06
		SAT-ML	91.83±0.15	7.69±0.26	87.08±0.70	10.35±0.15	76.63±0.38	12.75±0.04	63.90±0.22	15.31±0.22
		JoCoR-ML	90.54±0.41	7.36±0.05	82.24±0.47	10.44±0.14	69.43±0.33	12.92±0.30	59.71±0.48	17.68±0.14
		PAIR-Net (ours)	**93.43±0.22**	**7.28±0.22**	**88.76±0.61**	**9.93±0.19**	**81.34±0.29**	**11.90±0.20**	**72.95±0.31**	**14.93±0.21**
AID	Uniform	BCE	80.19±0.28	6.81±0.10	75.10±0.30	8.54±0.13	66.77±0.96	11.60±0.27	55.91±0.71	16.36±0.52
		RCML	79.94±0.14	6.59±0.12	75.38±0.55	8.10±0.10	67.40±0.32	11.10±0.15	57.66±1.62	15.54±0.68
		SAT-ML	81.60±0.30	6.65±0.19	77.99±0.52	**7.12±0.10**	74.21±0.50	**9.31±0.20**	69.45±0.24	14.59±0.51
		JoCoR-ML	81.50±0.31	6.66±0.05	76.74±0.87	8.63±0.18	70.02±0.58	11.88±0.24	59.74±0.56	16.43±0.19
		PAIR-Net (ours)	**82.53±0.65**	**6.48±0.17**	**78.93±0.65**	7.57±0.49	**75.36±0.41**	10.10±0.42	**71.64±0.58**	**14.54±0.15**
	Partial uniform	BCE	80.58±0.57	6.19±0.04	77.74±0.93	6.45±0.16	75.08±0.71	7.32±0.11	72.39±0.52	8.25±0.46
		RCML	82.02±0.33	**5.92±0.12**	79.03±0.87	6.33±0.15	76.13±0.38	7.05±0.10	73.93±0.36	7.80±0.62
		SAT-ML	80.85±0.52	6.29±0.07	78.34±0.64	6.48±0.14	75.66±0.27	**6.65±0.13**	73.86±0.25	**6.90±0.12**
		JoCoR-ML	80.08±0.41	6.12±0.12	78.14±0.63	6.36±0.21	74.94±0.52	7.36±0.12	71.76±0.64	8.47±0.61
		PAIR-Net (ours)	**82.64±0.53**	6.04±0.42	**79.82±1.35**	**6.27±0.23**	**77.64±0.63**	6.91±0.21	**74.01±0.63**	7.41±0.22
	Distribution-aware	BCE	80.43±0.61	8.23±0.37	73.63±0.38	11.79±0.26	65.03±0.20	16.08±0.47	57.88±0.86	20.59±0.16
		RCML	79.38±0.50	8.08±0.17	73.34±0.35	11.70±0.07	65.57±0.57	15.86±0.20	57.82±1.47	19.93±0.52
		SAT-ML	81.92±0.14	**7.63±0.11**	78.22±0.50	11.38±0.12	67.16±0.23	14.20±0.09	61.82±0.33	16.07±0.03
		JoCoR-ML	80.41±0.44	7.70±0.24	73.55±0.13	11.54±0.24	66.11±0.28	16.45±0.29	60.69±0.74	20.52±0.54
		PAIR-Net (ours)	**82.38±0.44**	7.69±0.50	**79.18±0.30**	**10.57±0.49**	**68.78±0.38**	**14.06±0.46**	**62.75±0.62**	**16.05±0.36**
	IDMN	BCE	76.34±0.49	9.30±0.04	69.88±0.07	13.34±0.05	63.90±0.36	17.96±0.17	55.94±0.26	23.53±0.06
		RCML	76.49±0.35	8.71±0.16	70.84±0.27	12.58±0.14	65.16±0.33	17.48±0.24	56.14±0.48	23.40±0.13
		SAT-ML	76.99±0.55	8.59±0.07	69.97±0.11	12.15±0.12	63.09±0.37	16.75±0.07	55.34±0.47	22.22±0.08
		JoCoR-ML	76.42±0.17	8.90±0.20	68.90±0.26	12.72±0.03	63.56±0.30	17.76±0.16	54.43±0.32	23.39±0.39
		PAIR-Net (ours)	**79.82±0.62**	**8.48±0.45**	**76.13±0.80**	**11.80±0.14**	**70.52±1.05**	**15.27±0.61**	**64.06±0.61**	**19.31±0.10**

tokens may introduce redundant prompt parameters and weaken the general semantic prior of CLIP under noisy supervision. To balance noise robustness and effective representation learning, we employ $M = 16$ for PAIR-Net in all experiments.

3) *Impact of Different Text Encoders:* We investigate the influence of CLIP text encoders in PAIR-Net on UCMerced and AID under mixed noise settings. The results in Table VII demonstrate that incorporating textual semantic information improves performance over the BCE baseline, indicating that text semantic priors are beneficial for robust RS multilabel classification under noisy supervision. Among the evaluated encoders, ViT-L/14 achieves the best overall performance across both datasets and noise rates. These results suggest that a stronger text encoder provides more discriminative semantic representations, enabling PAIR-Net to learn robust class-specific features under mixed noise settings. Thus, the text encoder of CLIP ViT-L/14 is employed in PAIR-Net.

4) *Impact of Different Numbers of Attention Heads:* We investigate the influence of the number of attention heads in the PAI module on PAIR-Net. As illustrated in Fig. 7, increasing the number of heads from 1 to 4 generally enhances classification performance on both UCMerced and AID datasets. This trend suggests that multiple

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TABLE V

IMPACT OF DIFFERENT COMPONENTS IN PAIR-NET ON THE UCMERCED (UCM) AND AID DATASETS UNDER MIXED NOISE SETTINGS

Dataset	Method	Mix-20%		Mix-40%	
		mAP↑	Ham.↓	mAP↑	Ham.↓
UCM	BCE	78.31±0.98	11.46±0.12	53.96±0.70	17.65±0.18
	+ PAI	82.15±0.72	11.36±0.28	58.36±0.73	16.41±0.17
	+ GradCAM	85.29±0.20	10.98±0.35	71.33±2.67	14.57±0.32
	+ ADF w/o $\mathcal{P}$	86.36±0.32	10.47±0.26	73.94±2.43	14.72±0.35
	PAIR-Net w/o PAI	83.84±0.51	10.95±0.36	68.91±2.64	15.42±0.33
	PAIR-Net	**87.42±0.48**	**10.16±0.09**	**75.53±0.60**	**14.41±0.06**
AID	BCE	68.96±0.09	14.44±0.28	54.74±0.43	24.37±0.09
	+ PAI	70.81±0.46	13.97±0.16	58.77±0.96	23.40±0.03
	+ GradCAM	71.11±0.72	12.48±0.12	61.21±0.32	19.88±0.12
	+ ADF w/o $\mathcal{P}$	72.59±0.78	12.52±0.16	62.75±0.54	19.82±0.10
	PAIR-Net w/o PAI	73.95±0.27	12.53±0.04	62.02±0.21	20.27±0.45
	PAIR-Net	**75.18±0.82**	**12.46±0.12**	**63.91±0.44**	**19.45±0.07**

TABLE VI

IMPACT OF LEARNABLE CONTEXT TOKENS M ON THE UCMERCED (UCM) AND AID DATASETS UNDER MIXED NOISE SETTINGS

Dataset	M	Mix-20%		Mix-40%	
		mAP↑	Ham.↓	mAP↑	Ham.↓
UCM	0	86.50±0.42	10.65±0.16	68.93±1.04	14.60±0.29
	4	86.15±1.40	10.59±0.31	75.30±0.71	14.60±0.28
	8	87.01±0.81	10.17±0.26	75.41±0.89	14.53±0.21
	16	87.42±0.48	**10.16±0.09**	**75.53±0.60**	**14.41±0.06**
	20	**87.65±0.75**	10.24±0.32	74.53±1.27	14.71±0.30
AID	0	72.57±1.32	12.58±0.14	59.22±0.11	19.72±0.48
	4	74.14±1.03	12.52±0.11	62.05±1.07	19.69±0.37
	8	74.46±0.99	12.50±0.12	63.13±1.97	19.53±0.32
	16	**75.18±0.82**	12.46±0.12	**63.91±0.44**	**19.45±0.07**
	20	74.56±1.21	**12.26±0.53**	62.82±1.26	19.57±0.29

TABLE VII

IMPACT OF DIFFERENT TEXT ENCODERS IN PAIR-NET ON UCMERCED (UCM) AND AID UNDER MIXED NOISE SETTINGS

Dataset	Text Encoder	Mix-20%		Mix-40%	
		mAP↑	Ham.↓	mAP↑	Ham.↓
UCM	BCE	78.31±0.98	11.46±0.12	53.96±0.70	17.65±0.18
	ViT-B/16	86.62±1.06	10.62±0.15	72.53±0.60	14.93±0.12
	ViT-B/32	87.04±0.58	10.73±0.19	73.90±0.62	14.86±0.19
	ViT-L/14	**87.42±0.48**	**10.16±0.09**	**75.53±0.60**	**14.41±0.06**
AID	BCE	68.96±0.09	14.44±0.28	54.74±0.43	24.37±0.09
	ViT-B/16	73.53±0.40	12.60±0.13	61.95±0.62	19.91±0.29
	ViT-B/32	74.18±0.85	12.53±0.06	63.41±0.13	19.71±0.38
	ViT-L/14	**75.18±0.82**	**12.46±0.12**	**63.91±0.44**	**19.45±0.07**

attention heads capture diverse cross-modal interaction patterns and strengthen RS feature discrimination under noisy supervision. However, further increasing the number of heads to 8 or 16 yields no consistent gains and even causes slight degradation in some cases. This phenomenon can be attributed to the fact that excessive attention heads introduce redundant interaction patterns and make the model more sensitive to noisy semantic correlations. Moreover, a larger number of heads inevitably increases the computational overhead. Thus, we set the number of attention heads to 4 for PAIR-Net in all experiments to balance classification performance and model efficiency.

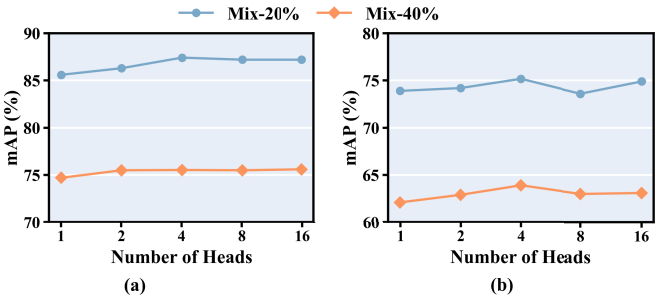


Fig. 7. Classification performance with different numbers of attention heads on (a) UCMerced and (b) AID datasets under mixed noise settings.

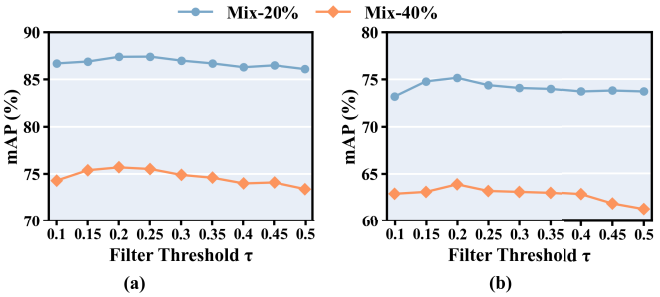


Fig. 8. Classification performance with different filtering thresholds τ on (a) UCMerced and (b) AID datasets under mixed noise settings.

5) *Impact of Different Filter Thresholds:* We further investigate the influence of the filter threshold τ within our ADF module. The parameter τ controls the tolerance to cross-class semantic interference, where a positive label is filtered if its **SCS** exceeds this threshold. Specifically, a smaller τ indicates a more aggressive filtering strategy, meaning that the model faces a greater risk of erroneously discarding genuine positive labels. Conversely, a larger τ implies conservative filtering, which struggles to detect and clean FP labels. As illustrated in Fig. 8, the performance shows a consistent favorable range across datasets and noise conditions. Specifically, mAP generally reaches its best value at $\tau = 0.2$ on both UCMerced and AID datasets. When τ is overly low (i.e., 0.1), the performance becomes suboptimal, reflecting the unintended removal of hard but clean present labels. Conversely, as τ increases beyond 0.3, the performance experiences noticeable degradation. This result demonstrates that an excessively high threshold fails to effectively suppress deceptive additive noise labels. Thus, to strike an optimal balance between preserving clean targets and eliminating noise labels, we set $\tau = 0.2$ as the filter threshold for all experiments.

6) *Impact of the Number of Warm-up Epochs:* The warm-up phase allows the model to develop sufficient RS visual representations before our dynamic filtering (ADF) is activated. As demonstrated in Fig. 9, using no warm-up

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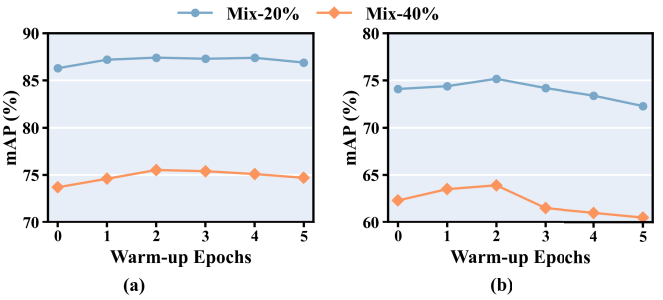


Fig. 9. Classification performance with different numbers of warm-up epochs on (a) UCMerced and (b) AID datasets under mixed noise settings.

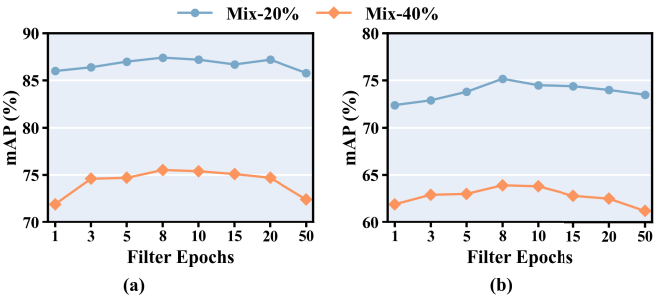


Fig. 10. Classification performance with different numbers of filtering epochs on (a) UCMerced and (b) AID datasets under mixed noise settings.

stage or only one warm-up epoch results in suboptimal performance. This result indicates that the model has not learned reliable activation maps without a sufficient warm-up stage, causing our ADF to mistakenly discard clean labels. The performance reaches its best value when the warm-up duration is set to two epochs, suggesting that the network has learned adequate preliminary features to generate reasonably reliable activations for filtering at this early training stage. However, further increasing the number of warm-up epochs degrades performance. This phenomenon can be attributed to the gradual memorization of noisy labels, which corrupts the activation maps and weakens the reliability of ADF. Thus, we adopt a two-epoch warm-up stage for PAIR-Net in all experiments to balance activation reliability against the risk of noise memorization.

7) *Impact of Different Filtering Durations:* We investigate the influence of the filtering epochs, which determine the duration of the dynamic noise filtering phase. As shown in Fig. 10, an insufficient number of filtering epochs, i.e., from one to five epochs, yields suboptimal performance. A short filtering phase fails to sufficiently purify the training set, leaving many noisy labels that continue to corrupt the learned representations. The performance peaks at eight filtering epochs, indicating that the filtering mechanism can effectively remove deceptive noisy labels within this duration while avoiding excessive correction of clean positive labels. However, further extending the filtering duration, i.e., from 15 to 50 epochs, slightly degrades performance. This phenomenon occurs because the model may gradually capture noise-specific patterns in later training stages, which leads to less reliable spatial activation maps and increases the risk of accidentally filtering clean samples. Thus, we set the number of filtering epochs to 8 to balance noise purification and training stability.

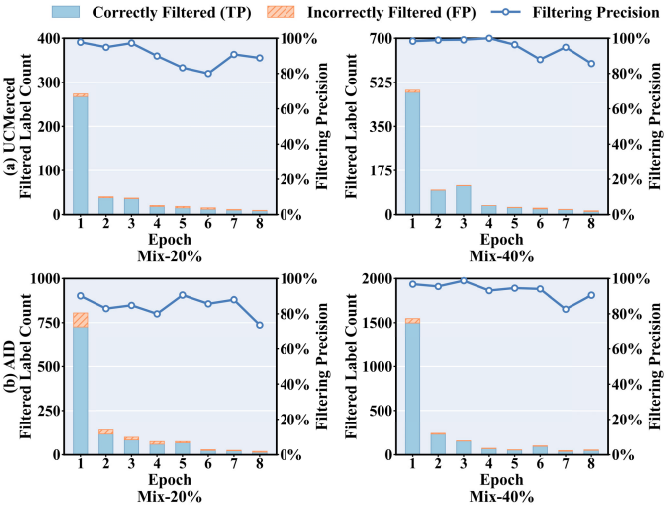


Fig. 11. Filtered label quantities and precision of PAIR-Net during the entire filtering stage under DD-IDMN on the UCMerced and AID datasets. The stacked bars report the numbers of correctly filtered additive noisy labels TP and incorrectly filtered genuine positive labels (FP), while the blue curves denote the filtering precision, i.e., TP/(TP + FP).



Fig. 12. Case study of noise identification under the Additive-60% DD-IDMN setting, where red labels indicate additive noisy labels. The filtered labels obtained by PAIR-Net show a clear reduction in additive noisy labels.

8) *Filtering Reliability Analysis:* To analyze PAIR-Net's filtering reliability, we record the number of filtered labels and filtering precision on UCMerced and AID. As shown in Fig. 11, PAIR-Net maintains high filtering precision, averaging 94.79% across evaluated settings. Moreover, most noisy labels are removed in early filtering epochs, while only a few are filtered in later epochs. This phenomenon suggests that PAIR-Net mainly removes highly suspicious positives at the early stage and avoids aggressively discarding clean positive labels.

To further examine filtering behavior under a more severe noise setting, we present a noise identification case study at Additive-60% in Fig. 12. This visualization further supports PAIR-Net's ability to reliably reduce additive noise under more challenging and diverse noisy supervision.

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TABLE VIII
COMPUTATIONAL COST COMPARISON ON THE MLRSNET DATASET WITH 60 OUTPUT CATEGORIES USING A SINGLE NVIDIA RTX 3090 GPU

Method	Params (M)	FLOPs (G)	Inference (ms/img)
BCE	23.63	4.13	0.70
PAIR-Net	32.17	4.63	0.78

J. Computational Efficiency

As summarized in Table VIII, we report the computational cost on MLRSNet with 60 output categories, including trainable parameters, FLOPs, and inference time. Compared with BCE, PAIR-Net incurs minor inference overhead, since the filtering module is used only during training. Thus, PAIR-Net improves noise robustness with limited inference latency.

V. LIMITATION

In our current formulation, PAIR-Net is primarily designed to mitigate additive noise rather than jointly correct additive and subtractive noise. This design is motivated by empirical analysis under DD-IDMN, where additive noise yields more severe degradation than subtractive noise. Accordingly, PAIR-Net focuses on identifying and rectifying unreliable positive labels caused by additive corruption, without explicitly recovering missing positive labels. Despite this limitation, PAIR-Net remains effective across noise settings. Specifically, our prompt-guided representation learning exploits textual semantics to enhance RS visual discriminability under noisy supervision, while ADF suppresses FP supervision. In future work, we will explore unified label rectification to simultaneously suppress FP labels and recover missed true labels for improved noise robustness.

VI. CONCLUSION

In this study, we revisit IDMN generation for RS MLC. We identify the cross-distribution thresholding bias in IDMN, where the score distribution of present labels is utilized to determine both additive and subtractive noise. To address this limitation, we propose DD-IDMN, which decouples the score distributions of present and absent labels for precise noise control. Through experimental analysis under DD-IDMN, we observe that additive noise yields more severe performance degradation than subtractive noise. Guided by this observation, we propose PAIR-Net from an activation-aware label rectification perspective. Specifically, it combines prompt-guided RS representation learning with dynamic filtering of unreliable positive labels. By analyzing activation differences between TP and FP labels, we design the SCS to identify additive noisy labels whose activations are largely contained within those of other positive categories. Comprehensive experiments demonstrate that PAIR-Net outperforms representative noise-robust multilabel methods across various noise generation methods.

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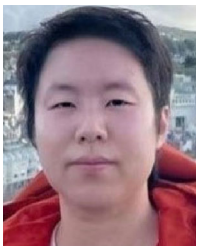
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